

A decorative border of various tropical leaves, including monstera and palm fronds, in shades of green, framing the central text.

Improper Food Storage

Grain Preservation

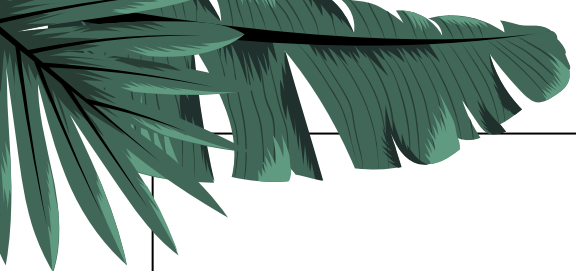


50–60%

of grain can be lost during
the storage stage

A decorative border of various tropical leaves, including monstera and palm fronds, in shades of green, framing the central text.

Evaporative Cooling



Demeter

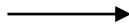
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Lewis Zhang, Ryan O'Connell

Process Explanation



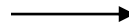
Air Input

Bringing Hot Air
Into Our System



Heat Transfer

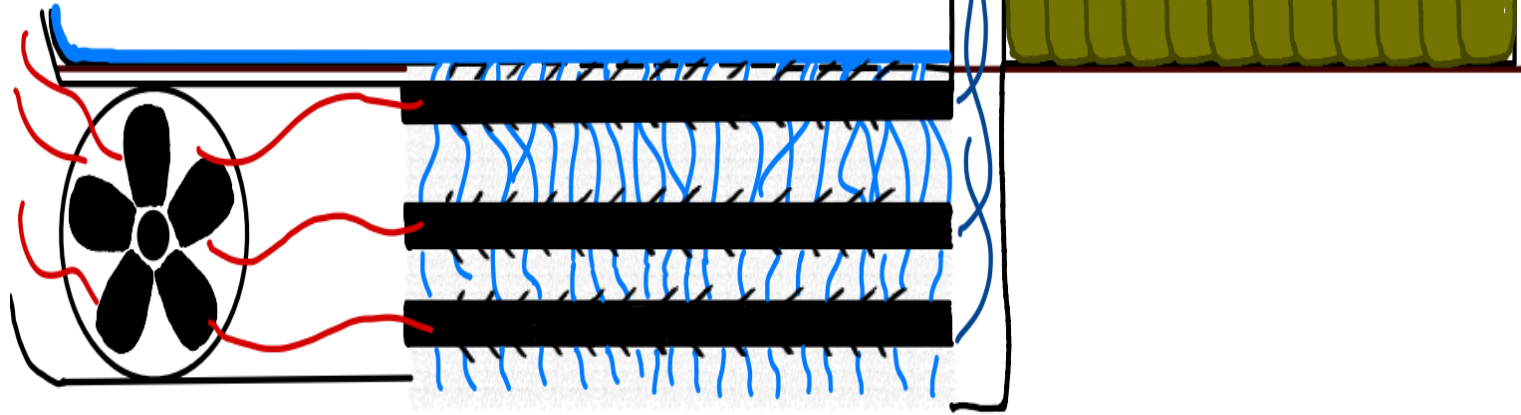
Pulling Heat
From the Air by
Evaporating
Water Around it



Cooling Food

The Newly
Cooled Air Moves
Towards Grain

The system leverages evaporative cooling to maintain a stable temperature, below 15°C [2].



Installing and maintaining grain bins
is extremely costly compared to
using grain bags and building a
shelter.



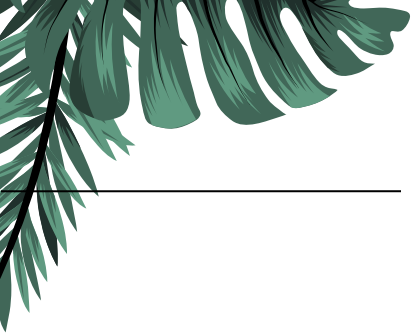
Grain bins

$\$50\,000 + \$48\,000/\text{yr} [3]$



Bags & Shelter

$< \$35\,000 + \$500/\text{yr}$



**Demeter allows McMasterton
farmers to cost-effectively extend
the storage life of their grain.**

**2 ZERO
HUNGER**



To end hunger, achieve food
security and improved nutrition,
and promote sustainable agriculture



References

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2. Grain storage [Internet]. [cited 2025 Jan 26]. Available from: <https://www.siousteel.com/explore-our-products/grain-storage>
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